Exam III Review: Sections 11.8-11.11, 10.1, & 10.5

Sections 11.8-11.9: Power Series & Functions as Power Series

1. Find the interval and radius of convergence for the given power series.

   (a) $\sum_{n=0}^{\infty} \frac{(x+2)^{2n+1}}{4^n n^4}$ (b) $\sum_{n=0}^{\infty} \frac{(n+1)!(x-5)^n}{n^2+2n}$ (c) $\sum_{n=0}^{\infty} \frac{7^n(3x-1)^n}{3\cdot 5\cdot 7\cdots(2n+3)}$

2. Find the power series representation for the following functions. Determine the interval and radius of convergence.

   (a) $f(x) = \frac{2}{4+x^2}$ (b) $f(x) = \frac{4}{(1+x^3)^2}$ (c) $f(x) = \ln(x^3+8)$

Sections 11.10-11.11: Taylor and MacLaurin Series & Applications

1. Use series to evaluate the limit: $\lim_{x\to 0} \frac{e^{x^3}-1}{\sin x - x}$.
2. Find the MacLaurin series of $f(x) = 4x^2 e^{4x^2}$ using a known series.
3. Approximate the definite integral $\int_0^{0.2} \frac{\tan^{-1}(x^4)}{4x^4}dx$ using series to find the first 3 nonzero terms.
4. Find the sum of the series: $\sum_{n=0}^{\infty} \frac{(-1)^n}{(\sqrt{3})^{2n+1}(2n+1)}$.
5. Use Binomial Theorem to find $f(x) = \frac{2x}{(x+2)^2}$ as a power series.
6. Find the Taylor series for $f(x) = \sqrt{2x-1}$ at $a = 13$.
7. Use Taylor's Inequality given below to approximate the function $f(x) = \ln(4x+5)$ by a Taylor polynomial of degree 3 at $a = 2$. How accurate is this approximation when $1.5 < x < 2.5$?

   If $\left|f^{(n+1)}(x)\right| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{(n+1)}$ for $|x-a| \le d$.

Section 10.1: Parametric Equations

1. Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases.

   (a) $x = 4\cos t$, $y = 2\sin t$, $0 \le t \le \frac{\pi}{2}$ (b) $x = 4-2t$, $y = 3+6t-t^2$, $0 \le t \le 3$

Section 10.5: Conic Sections

1. Find the equation of a parabola with vertex at (2,0) and directrix $x = -2$.
2. Find the equation of a parabola with focus at (2,6) and vertex at (2,3).
3. Find the equation of an ellipse with foci (-2,0), (4,0) and vertex (6,0).
4. Find the equation of a hyperbola with the vertices (-2,-1), (-2,1) and foci $(-2,\sqrt{5}), (-2,-\sqrt{5})$.
5. Find the vertex, focus, and directrix of $y^2 = 4x+4y-8$. Sketch the graph.
6. Find the center, foci, and vertices of $x^2+4y^2-4x+24y+39 = 0$. Sketch the graph.

7. Find the vertices, foci, asymptotes, and center of $4x^2 - y^2 + 8x + 2y - 13 = 0$. Sketch the graph.

True/False

1. Expanding $\sum_{n=0}^{\infty} \frac{(x+3)^{n+2}}{n+2^n}$ to the third degree would give $\frac{(x+3)^2}{1} + \frac{(x+3)^3}{1+2} + \frac{(x+3)^4}{2+2^2} + \frac{(x+3)^5}{3+2^3}$.
2. If $-4 < x < 5$, then $|x-2| < 6$.
3. The Taylor Series for $f(x) = e^x$ at $a = 1$ is given by $\sum_{n=0}^{\infty} \frac{(x-1)^n}{n!}$.
4. The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ is $\frac{\pi}{2}$.
5. The curve given by the parametric equations $x = 3\sin t$ and $y = 3\cos t$ describes a circle with a center at (0,0) with radius 3 that travels counterclockwise.
6. The directrix for the parabola $y^2 - 4y - 8x = 4$ is $y = -3$.

Multiple Choice

1. The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(x+2)^{2n}}{4^n}$ is

   (a) $\frac{1}{2}$ (b) 4 (c) 2 (d) $\frac{1}{4}$

2. The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x+1)^n}{n}$ is

   (a) $[-2, 0]$ (b) $(-2, 0)$ (c) $[-2, 0)$ (d) $(-2, 0]$

3. The derivative of the series $\sum_{n=0}^{\infty} n^2(x+4)^{2n+1}$ is

   (a) $\sum_{n=0}^{\infty} \frac{n^2(x+4)^{2n+2}}{2n+2}$
   (b) $\sum_{n=1}^{\infty} n^2(2n+1)(x+4)^{2n}$
   (c) $\sum_{n=0}^{\infty} n^2(2n+1)(x+4)^{2n}$
   (d) $\sum_{n=1}^{\infty} \frac{n^2(x+4)^{2n+2}}{2n+2}$

4. The series for $f(x) = x^2 \arctan(x^2)$ is

   (a) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+3}}{2n+1}$ (b) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$ (c) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+4}}{2n+1}$ (d) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+4}}{(2n+1)!}$

5. If $f(a) = 1$, $f'(a) = \frac{4}{3}$, $f''(a) = \frac{4 \cdot 7}{9}$, and $f'''(a) = \frac{4 \cdot 7 \cdot 10}{27}$, then

   (a) $f^{(n)}(a) = \frac{4 \cdot 7 \cdot 10 \cdot 13 \cdots (3n+4)}{3^{n+1}}$
   (b) $f^{(n)}(a) = \frac{4 \cdot 7 \cdot 10 \cdot 13 \cdots (3n+1)}{3^n}$
   (c) $f^{(n)}(a) = \frac{4 \cdot 7 \cdot 10 \cdot 13 \cdots (3n+1)}{3^{n+1}}$
   (d) $f^{(n)}(a) = \frac{4 \cdot 7 \cdot 10 \cdot 13 \cdots (3n+4)}{3^n}$

6. Which of the following is a focus for the ellipse $\frac{(x+1)^2}{9} + \frac{(y-2)^2}{25} = 1$?

   (a) (-1,2) (b) (-5,2) (c) (3,2) (d) (-1,6)

Exam III Review

Sections 11.8-11.9

1. (a) $\sum_{n=0}^{\infty} \frac{(x+2)^{2n+1}}{4^n n^4}$ Ratio Test

$$\lim_{n\to\infty}\left|\frac{(x+2)^2\cancel{(x+2)^{2n+1}}}{4\cdot\cancel{4^n}(n+1)^4}\cdot\frac{\cancel{4^n}\,n^4}{\cancel{(x+2)^{2n+1}}}\right| = \lim_{n\to\infty}\left|\frac{(x+2)^2 n^2}{4(n+1)^2}\right|$$

$$= \left|\frac{(x+2)^2}{4}\right| < 1$$

$$|(x+2)^2| < 4$$

$$|x+2| < \boxed{2 = R}$$

$$-2 < x+2 < 2$$

$$-4 < x < 0 \quad \boxed{I = [-4, 0]}$$

At $x=-4$: $\sum_{n=0}^{\infty} \frac{(-4+2)^{2n+1}}{4^n n^4} = \sum_{n=0}^{\infty} \frac{(-2)^{2n+1}}{4^n n^4}$

$$= \sum_{n=0}^{\infty} \frac{-2(-2)^{2n}}{4^n n^4}$$

$$= \sum_{n=0}^{\infty} \frac{-2\cdot\cancel{4^n}}{\cancel{4^n} n^4}$$

$$= \sum_{n=0}^{\infty} \frac{-2}{n^4}$$ Convergent by LCT

At $x=0$: $\sum_{n=0}^{\infty} \frac{(0+2)^{2n+1}}{4^n n^4}$

$$= \sum_{n=0}^{\infty} \frac{2}{n^4}$$

Convergent by LCT

(b) $\sum_{n=0}^{\infty} \frac{(n+1)!\,(x-5)^n}{n^2+2n}$ Ratio Test

$$\lim_{n\to\infty}\left|\frac{(n+2)\cancel{(n+1)!}\,(x-5)\cancel{(x-5)^n}}{(n+1)^2+2(n+1)}\cdot\frac{n^2+2n}{\cancel{(n+1)!}\,\cancel{(x-5)^n}}\right|$$

$$= \lim_{n\to\infty}\left|\frac{(n+2)(n^2+2n)(x-5)}{(n+1)^2+2(n+1)}\right| = \infty \not< 1$$

$$\boxed{R = 0,\ I = \{5\}}$$

(c) $\sum_{n=0}^{\infty} \frac{7^n(3x-1)^n}{3\cdot5\cdot7\cdots(2n+3)}$ Ratio Test

$$\lim_{n\to\infty}\left|\frac{7\cdot\cancel{7^n}(3x-1)\cancel{(3x-1)^n}}{\cancel{3\cdot5\cdot7\cdots(2n+3)}(2n+5)}\cdot\frac{\cancel{3\cdot5\cdot7\cdots(2n+3)}}{\cancel{7^n}\cancel{(3x-1)^n}}\right|$$

$$= \lim_{n\to\infty}\left|\frac{7(3x-1)}{2n+5}\right| = 0 < 1$$ Always $\boxed{R = \infty,\ I = (-\infty, \infty)}$

2. (a) $f(x) = \frac{2}{4+x^2} \cdot \frac{1/4}{1/4}$

$= \frac{1/2}{1-(-x^2/4)}$ $\quad a = 1/2,\ r = -\frac{x^2}{4}$

$= \sum_{n=0}^{\infty} \frac{1}{2}\left(\frac{-x^2}{4}\right)^n$ $\quad |r| = \left|\frac{-x^2}{4}\right| < 1$

$|x^2| < 4$

$|x| < 2$ $\quad \boxed{R = 2,\ I = (-2, 2)}$

$= \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n+1}}}$

(b) $f(x) = \frac{4}{(1+x^3)^2}$ ← We need to integrate this to get in the form $\frac{a}{1-r}$, but how do you integrate $\int \frac{4}{(1+x^3)^2}\,dx$? There's no $3x^2$ for the du.

So let's consider a temporary substitution:

$|r| = |-m|$

$= |-x^3| < 1$

$-1 < x < 1$

$\boxed{R = 1,\ I = (-1, 1)}$

$\frac{4}{(1+m)^2} = \frac{d}{dm}\left(\int 4(1+m)^{-2}\,dm\right)$

$= \frac{d}{dm}\left(-4(1+m)^{-1} + c\right)$

$= \frac{d}{dm}\left(\frac{-4}{1+m} + c\right)$ $\quad a = -4,\ r = -m$

$= \frac{d}{dm}\left(\sum_{n=0}^{\infty} -4(-m)^n + c\right)$

$= \frac{d}{dm}\left(\sum_{n=0}^{\infty} 4(-1)^{n+1} m^n + c\right)$

$= \sum_{n=1}^{\infty} 4(-1)^{n+1} n\, m^{n-1}$

So $f(x) = \sum_{n=1}^{\infty} 4(-1)^{n+1} n\,(x^3)^{n-1} = \boxed{\sum_{n=1}^{\infty} 4(-1)^{n+1} n\, x^{3n-3}}$

* Note this substitution only works if we go back to x after integrating and then undoing the integration (taking the derivative)

(c) $f(x) = \ln(x^3+8)$ ← Need to differentiate to get in the form $\frac{a}{1-r}$

$$= \int \frac{d}{dx}\left(\ln(x^3+8)\right) dx$$

$$= \int \frac{3x^2}{x^3+8}\, dx$$

$$= \int \frac{3x^2/8}{1-(-x^3/8)}\, dx \qquad a = \frac{3x^2}{8},\ r = \frac{-x^3}{8}$$

$$|r| = \left|\frac{-x^3}{8}\right| < 1$$

$$|x^3| < 8$$

$$|x| < \boxed{2 = R,\ I = (-2,2)}$$

$$= \int \sum_{n=0}^{\infty} \left(\frac{3x^2}{8}\right)\left(\frac{-x^3}{8}\right)^n dx$$

$$= \int \sum_{n=0}^{\infty} \frac{3(-1)^n x^{3n+2}}{8^{n+1}}\, dx$$

$$= \boxed{\sum_{n=0}^{\infty} \frac{3(-1)^n x^{3n+3}}{8^{n+1}(3n+3)} + C}$$

* Note: solving for c is not necessary, but if you want to know how:

$$f(0) = \ln(0^3+8) = \sum_{n=0}^{\infty} \frac{3(-1)^n 0^{3n+3}}{8^{n+1}(3n+3)} + C$$

$$\ln 8 = C$$

Sections 11.10 – 11.11

1. $$\lim_{x\to 0} \frac{e^{x^3}-1}{\sin x - x} = \lim_{x\to 0} \frac{-1 + \sum_{n=0}^{\infty} \frac{x^{3n}}{n!}}{-x + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}}$$

$$= \lim_{x\to 0} \frac{-1 + \frac{1}{1} + \frac{x^3}{1} + \frac{x^6}{2!} + \frac{x^9}{3!} + \cdots}{-x + x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots}$$

$$= \lim_{x\to 0} \frac{x^3 + \frac{x^6}{2!} + \frac{x^9}{3!} + \cdots}{-\frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots}$$

$$= \lim_{x\to 0} \frac{\cancel{x^3}\left(1 + \frac{x^3}{2!} + \frac{x^6}{3!} + \cdots\right)}{\cancel{x^3}\left(-\frac{1}{3!} + \frac{x^2}{5!} - \frac{x^4}{7!} + \cdots\right)} = \frac{1}{-1/3!} = \boxed{-6}$$

2. $f(x) = 4x^2 e^{4x^2}$

$$= 4x^2 \sum_{n=0}^{\infty} \frac{(4x^2)^n}{n!}$$

$$= \boxed{\sum_{n=0}^{\infty} \frac{4^{n+1} x^{2n+2}}{n!}}$$

3. $$\int_0^{0.2} \frac{\tan^{-1}(x^4)}{4x^4}\,dx = \int_0^{0.2} \frac{1}{4x^4} \sum_{n=0}^{\infty} \frac{(-1)^n (x^4)^{2n+1}}{2n+1}\,dx$$

$$= \int_0^{0.2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{8n}}{4(2n+1)}\,dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{8n+1}}{4(2n+1)(8n+1)} \Bigg|_0^{0.2}$$

$$\approx \frac{x}{4(2)(8)} - \frac{x^9}{4(3)(9)} + \frac{x^{17}}{4(5)(17)} \Bigg|_0^{0.2}$$

$$= \boxed{\frac{0.2}{4(2)(8)} - \frac{0.2^9}{4(3)(9)} + \frac{0.2^{17}}{4(5)(17)} - 0}$$

4. $$\sum_{n=0}^{\infty} \frac{(-1)^n}{(\sqrt{3})^{2n+1}(2n+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n (1/\sqrt{3})^{2n+1}}{2n+1}$$

$$= \arctan\left(\frac{1}{\sqrt{3}}\right) = \boxed{\pi/6}$$

5. $$f(x) = \frac{2x}{(x+2)^2} = 2x\left(2(1+x/2)\right)^{-2} \quad K=-2$$

$$= \frac{2x}{2^2}(1+x/2)^{-2}$$

$$= \frac{x}{2}(1+x/2)^{-2}$$

$$= \frac{x}{2} \sum_{n=0}^{\infty} \square \frac{(x/2)^n}{n!}$$

$1,\ K,\ K(K-1),\ K(K-1)(K-2),\ K(K-1)(K-2)(K-3), \ldots$

$1, -2, -2(-3), -2(-3)(-4), -2(-3)(-4)(-5), \ldots$

$1, -2, 2(3), -2(3)(4), 2(3)(4)(5), \ldots$

$$\underset{n=0}{1},\ \underset{n=1}{-2!},\ \underset{n=2}{3!},\ \underset{n=3}{-4!},\ \underset{n=4}{5!},\ \ldots$$

$$\boxed{} = (-1)^n (n+1)! \leftarrow \text{pattern}$$

So $$f(x) = \frac{x}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (n+1)! \left(\frac{x}{2}\right)^n}{n!}$$

$$= \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n (n+1)\, x^{n+1}}{2^{n+1}}}$$

6. $f(x) = \sqrt{2x-1}$ $\quad a = 13$

$$f(x) = (2x-1)^{1/2} \qquad f(13) = 5$$

$$f'(x) = \frac{1}{2}(2x-1)^{-1/2} \cdot 2 \qquad f'(13) = \frac{1}{5}$$

$$f''(x) = \frac{-1}{2^2}(2x-1)^{-3/2} \cdot 2^2 \qquad f''(13) = \frac{-1}{5^3}$$

$$f'''(x) = \frac{1 \cdot 3}{2^3}(2x-1)^{-5/2} \cdot 2^3 \qquad f'''(13) = \frac{1 \cdot 3}{5^5}$$

$$f''''(x) = \frac{-1 \cdot 3 \cdot 5}{2^4}(2x-1)^{-7/2} \cdot 2^4 \qquad f''''(13) = \frac{-1 \cdot 3 \cdot 5}{5^7}$$

$$\underset{n=0}{5},\ \underset{n=1}{\frac{1}{5}},\ \underset{n=2}{\frac{-1}{5^3}},\ \underset{n=3}{\frac{1 \cdot 3}{5^5}},\ \underset{n=4}{\frac{-1 \cdot 3 \cdot 5}{5^7}},\ \ldots$$

pattern: $$f^{(n)}(13) = \frac{(-1)^{n+1}\, 1 \cdot 3 \cdot 5 \cdots (2n-3)}{5^{2n-1}}$$

will not work for $n=0$, $n=1$ since it will be negative

$$f(x) = \frac{f(13)(x-13)^0}{0!} + \frac{f'(13)(x-13)^1}{1!} + \sum_{n=2}^{\infty} \frac{f^{(n)}(13)(x-13)^n}{n!}$$

$$= \boxed{5 + \frac{1}{5}(x-13) + \sum_{n=2}^{\infty} \frac{(-1)^{n+1}(1 \cdot 3 \cdot 5 \cdots (2n-3))(x-13)^n}{5^{2n-1}\, n!}}$$

7. $f(x) = \ln(4x+5)$, T_3 at $a=2$

$f(x) = \ln(4x+5)$ $\qquad f(2) = \ln(13)$

$f'(x) = \frac{4}{4x+5}$ $\qquad f'(2) = \frac{4}{13}$

$f''(x) = \frac{-4^2}{(4x+5)^2}$ $\qquad f''(2) = \frac{-16}{13^2}$

$f'''(x) = \frac{2 \cdot 4^3}{(4x+5)^3}$ $\qquad f'''(2) = \frac{128}{13^3}$

$f''''(x) = \frac{-6 \cdot 4^4}{(4x+5)^4}$

$$T_3(x) = \frac{f(2)(x-2)^0}{0!} + \frac{f'(2)(x-2)^1}{1!} + \frac{f''(2)(x-2)^2}{2!} + \frac{f'''(2)(x-2)^3}{3!}$$

$$\boxed{T_3(x) = \ln 13 + \frac{4}{13}(x-2) - \frac{16}{13^2 \cdot 2}(x-2)^2 + \frac{128}{13^3 \cdot 6}(x-2)^3}$$

For Taylor Inequality:

$$R_n \le \frac{M}{(n+1)!} d^{n+1}$$

$M: |f''''(x)| = \left| \frac{-6 \cdot 4^4}{(4x+5)^4} \right|$ $\qquad$ on $1.5 < x < 2.5$

$$\le \frac{6 \cdot 4^4}{[4(1.5)+5]^4} = M$$

$$\boxed{R_3 \le \frac{6 \cdot 4^4}{[4(1.5)+5]^4}\left(\frac{0.5^4}{4!}\right)}$$

$d: 1.5 < x < 2.5$

$1.5 - 2 < x - 2 < 2.5 - 2$

$-0.5 < x - 2 < 0.5$

$|x-2| < 0.5 = d$

Section 10.1

1.(a) $x = 4\cos t$, $y = 2\sin t$ $\quad 0 \le t \le \pi/2$

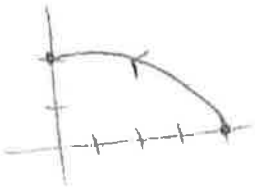
$\cos t = \frac{x}{4}$ $\quad \sin t = \frac{y}{2}$

$\cos^2 t + \sin^2 t = 1$ (Identity!)

$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$

$$\boxed{\frac{x^2}{16} + \frac{y^2}{4} = 1}$$

That's an ellipse! going counter clockwise and $0 \le t \le \pi/2$ means a fourth of the ellipse starting at $(4,0)$ (when $t=0$) and ending at $(0,2)$ (when $t = \pi/2$)



(b) $x = 4 - 2t$, $y = 3 + 6t - t^2$, $0 \le t \le 3$

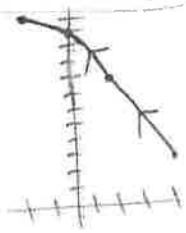
$t = 2 - \frac{x}{2}$

$y = 3 + 6\left(2 - \frac{x}{2}\right) - \left(2 - \frac{x}{2}\right)^2$

$= 3 + 12 - 3x - 4 + 2x - \frac{x^2}{4}$

$$\boxed{y = 11 - x - \frac{x^2}{4}}$$

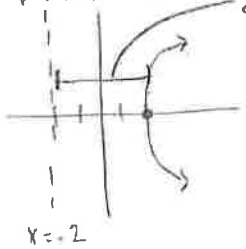
t	x	y
0	4	3
1	2	8
2	0	11
3	-2	12



Section 10.5

1. vertex at $(2,0)$, directrix at $x = -2$

$p = 4$

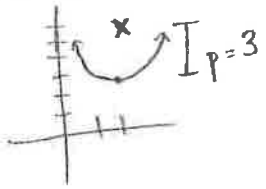


$(y-k)^2 = 4p(x-h)$

$(y-0)^2 = 4(4)(x+2)$

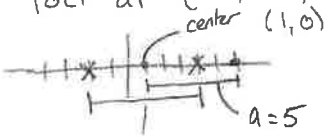
$$\boxed{y^2 = 16(x+2)}$$

2. focus at $(2,6)$, vertex at $(2,3)$



$$(x-h)^2 = 4p(y-h)$$
$$(x-2)^2 = 4(3)(y-3)$$
$$\boxed{(x-2)^2 = 12(y-3)}$$

3. foci at $(-2,0)$, $(4,0)$, vertex at $(6,0)$

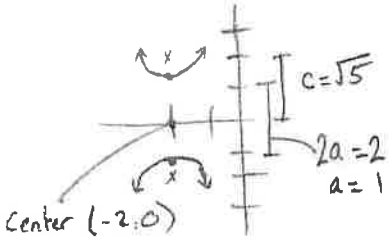


$2c = 6$
$c = 3$

$b^2 = a^2 - c^2$
$= 25 - 9 = 16$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$
$$\boxed{\frac{(x-1)^2}{25} + \frac{(y-0)^2}{16} = 1}$$

4. vertices $(-2,-1)$, $(-2,1)$, foci $(-2,\sqrt{5})$, $(-2,-\sqrt{5})$



$b^2 = c^2 - a^2 = 5 - 1 = 4$

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$
$$\boxed{\frac{(y-0)^2}{1} - \frac{(x+2)^2}{4} = 1}$$

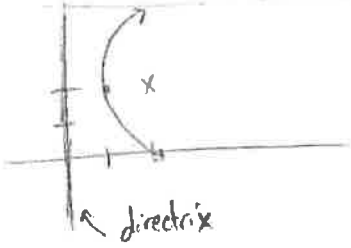
5. $y^2 = 4x + 4y - 8$

$y^2 - 4y + 4 = 4x - 8 + 4$

$(y-2)^2 = \underbrace{4}_{\text{focal width} = 4p = 4}(x-1)$

$p = 1$

vertex : $(1,2)$
focus : $(1+p, 2) = (2,2)$
directrix : $x = 1 - p = 0$



6. $x^2 + 4y^2 - 4x + 24y + 39 = 0$

$x^2 - 4x + 4 + 4(y^2 + 6y + 9) = -39 + 4 + 4(9)$

$(x-2)^2 + 4(y+3)^2 = 1$

$\frac{(x-2)^2}{1} + \frac{(y+3)^2}{1/4} = 1$

$a^2 = 1 \Rightarrow a = 1$

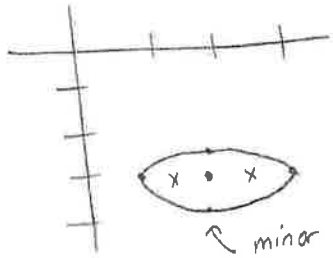
$b^2 = 1/4 \Rightarrow b = 1/2$

$c^2 = a^2 - b^2 = 1 - 1/4 = 3/4 \Rightarrow c = \sqrt{3}/2$

Center: $(2, -3)$

foci: $(2 \pm c, -3) = (2 \pm \sqrt{3}/2, -3)$

vertices: $(2 \pm a, -3) = (3, -3), (1, -3)$



minor axis vertices are $(2, -3 \pm b)$

7. $4x^2 - y^2 + 8x + 2y - 13 = 0$

$4(x^2 + 2x + 1) - (y^2 - 2y + 1) = 13 + 4(1) + (-1)(1)$

$4(x+1)^2 - (y-1)^2 = 16$

$\frac{(x+1)^2}{4} - \frac{(y-1)^2}{16} = 1$

$a^2 = 4 \Rightarrow a = 2$

$b^2 = 16 \Rightarrow b = 4$

$c^2 = a^2 + b^2 = 20 \Rightarrow c = \sqrt{20}$

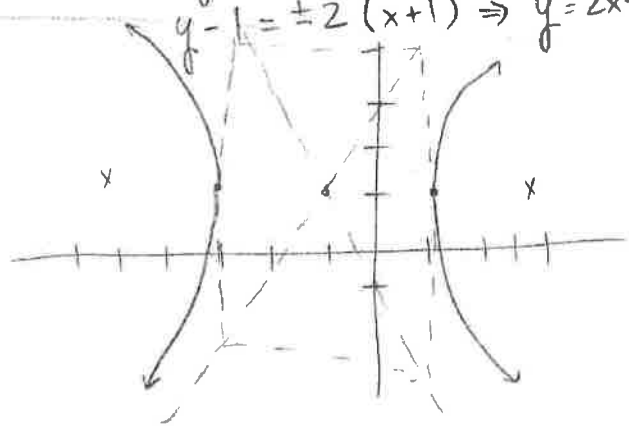
Center: $(-1, 1)$

vertices: $(-1 \pm a, 1) = (-3, 1), (1, 1)$

foci: $(-1 \pm c, 1) = (-1 \pm \sqrt{20}, 1)$

asymptotes: $y - k = \pm \frac{b}{a}(x - h)$

$y - 1 = \pm 2(x+1) \Rightarrow y = 2x + 3,\ y = -2x - 1$



True / False

1. $\boxed{\text{False}}$, third degree means until $(x-a)^3$

2. $-4 < x < 5$

$-4-2 < x-2 < 5-2$

$-6 < x-2 < 3$

bigger Number so $|x-2| < 6$ $\boxed{\text{True}}$

3. $\boxed{\text{False}}$ Taylor Series has the formula $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)\,(x-a)^n}{n!}$

$f(x) = e^x \Rightarrow f^{(n)}(x) = e^x \Rightarrow f^{(n)}(1) = e$

but there is no e in that formula

4. $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n\, 1^{2n+1}}{2n+1} = \arctan(1) = \pi/4$ $\boxed{\text{False}}$

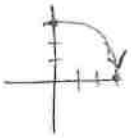
5. $x = 3\sin t$, $y = 3\cos t$

$\cos^2 t + \sin^2 t = 1 \Rightarrow \frac{y^2}{9} + \frac{x^2}{9} = 1$, so circle with center at $(0,0)$ with radius 3

What about direction? When $t=0$, $(0,3)$

$t = \pi/2$, $(3,0)$

clockwise!

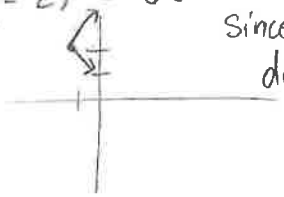


$\boxed{\text{False}}$

6. $y^2 - 4y - 8x = 4$

$y^2 - 4y + 4 = 8x + 4 + 4$

$(y-2)^2 = 8(x+1)$



Since it opens to the side, directrix will be $x =$ something $\boxed{\text{False}}$

Multiple Choice

1. Geometric Series : $|r| = \left|\frac{(x+2)^2}{4}\right| < 1$

$|(x+2)^2| < 4$

$|x+2| < 2 = R$ (c)

2. Ratio Test : $\lim_{n\to\infty} \left| \frac{(x+1)^{n+1}}{n+1} \cdot \frac{n}{\cancel{(x+1)^n}} \right| = |x+1| < 1$

$-1 < x+1 < 1$

$-2 < x < 1$

at $x=-2$: $\sum_{n=0}^{\infty} \frac{(-1)^n}{n}$ Convergent by AST

at $x=0$: $\sum_{n=0}^{\infty} \frac{1^n}{n}$ divergent p-series

$[2, 0)$ (c)

3. $\frac{d}{dx}\left(\sum_{n=0}^{\infty} n^2 (x+4)^{2n+1}\right) = \sum_{n=0}^{\infty} n^2 (2n+1)(x+4)^{2n}$ (c)

↑ do not shift the index

4. $f(x) = x^2 \arctan(x^2) = x^2 \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+4}}{2n+1}$ (c)

5. $n=3$, $f'''(a) = \frac{4\cdot 7\cdot 10}{3^3}$ $4\cdot 7\cdot 10 \cdots (3n+1)$ (b)

3^n

6. $\frac{(x+1)^2}{9} + \frac{(y-2)^2}{25} = 1$

$a^2 = 25$, $b^2 = 9$

$c^2 = 25 - 9 = 16$

$c = 4$

center : $(-1, 2)$

foci : $(-1, 2 \pm c)$

$(-1, 6)$, $(-1, -2)$

(d)